

TravelTide Customer Loyalty Perk Optimization

Masterschool Data Science & Analytics Mastery Project

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Agenda



1. Business Context
2. Data Overview
3. Customer Behavior Insights
4. Customer Segmentation
5. Dashboard Findings
6. Strategic Recommendations
7. Future Opportunities

Business Context



TravelTide

- Online travel platform
- Flights
- Hotels
- Travel packages

Business Challenge

How can TravelTide:

- Increase engagement?
- Reduce churn?
- Improve loyalty program effectiveness?

Project Objective

Identify the most effective loyalty perk for each customer segment.

Data & Methodology



Data

- 50,000 customers
- 105,000 flights
- 105,000 hotel bookings
- Session activity

Tools

- SQL
- Python
- Tableau

Features Created

- Age Segment
- Travel Frequency
- Hotel Usage
- Family Status
- Preferred Perk

Customer Behavior Insights



Insight #1

Customers Aged 40 - 49 Represent Highest-Value Segment

HIGHEST VALUE SEGMENT

40 - 49

30.8% of engaged customers

Customers with higher browsing activity are significantly more likely to complete bookings.

Customer Behavior Insights



Insight #2

Medium-frequency travelers represent the largest active segment (~43% of customers).

Medium-Frequency Travelers Represent the Largest Customer Segment



Medium-frequency travelers represent the largest engaged customer group and the strongest opportunity for loyalty campaigns.

Customer Behavior Insights



Insight #3

Customers booking both flights and hotels show 74% higher engagement than single-service customers.

Multi-Service Customers Show Stronger Engagement

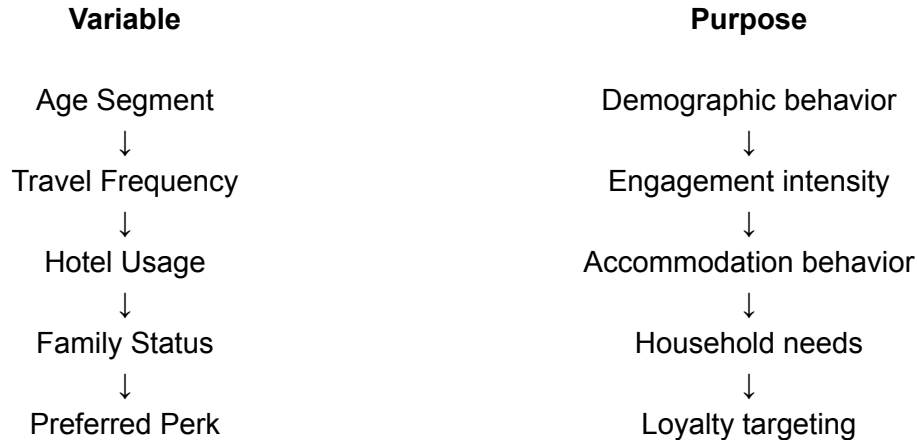


Customers using both hotels and flights demonstrate stronger engagement than customers using only one service.

Customer Segmentation Approach



Segmentation Variables Used



Goal: Create actionable segments for targeted loyalty campaigns.

Customer Segmentation Logic



Rule-Based Customer Segmentation Framework

Variable	Rule
Travel Frequency	Low / Medium / High
Age	18-29, 30-39, 40-49, 50-59, 60+
Hotel Usage	Low / Medium / High
Perk Preference	Highest Perk Score

Customers were grouped using behavioral and demographic features to identify the most effective loyalty incentive for each segment.

Segmentation Outcome



Rule-Based Segmentation

What We Learned

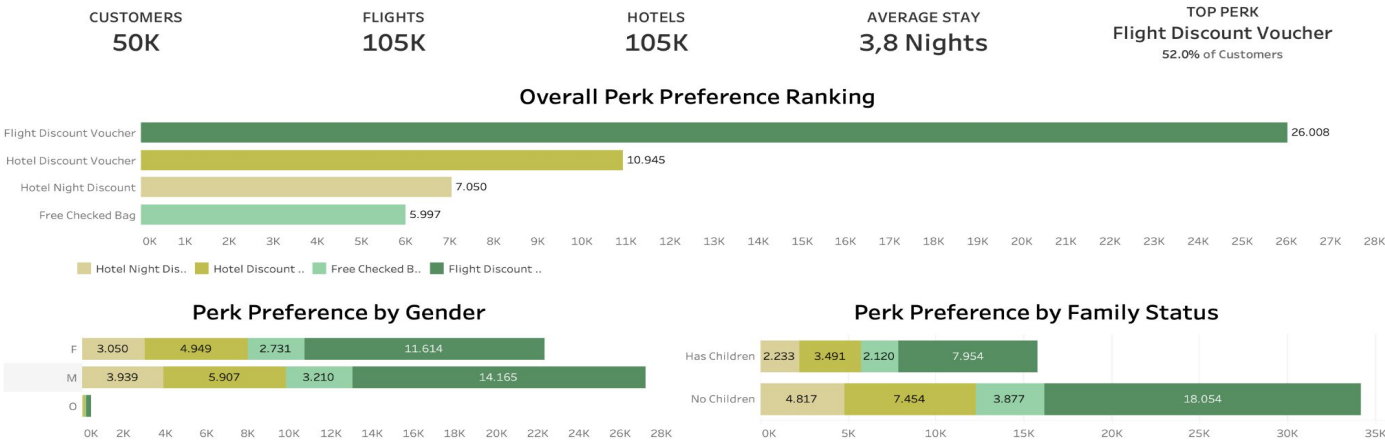
- Flight Discount Voucher preferred by 52%
- Age 40–49 highest-value segment
- Medium-frequency travelers largest engaged segment
- High hotel users show strongest preference for Free Checked Bag

Executive Overview Dashboard



TravelTide Customer Perks Recommendation Dashboard

Customer Segmentation and Loyalty Perk Optimization Analysis



KEY INSIGHT

Flight Discount Voucher is preferred by 52% of customers and remains the top loyalty incentive across gender and family segments, making it the strongest candidate for a standardized loyalty program.

Source: TravelTide Customer Dataset (50,000 Customers)

Tools Used: SQL • Python • Tableau

Prepared for: Masterschool Data Science & Analytics Program

Flight Discount Voucher is preferred by 52% of customers and consistently ranks first across demographic segments.

Customer Segment Insights Dashboard



Customer Behavior & Segment Analysis

Customer segmentation analysis to identify the most valuable loyalty campaign opportunities

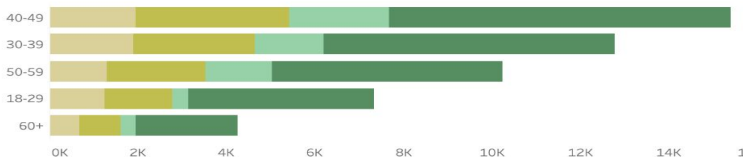
HIGHEST VALUE SEGMENT
40 - 49
30.8% of engaged customers

LARGEST TRAVEL SEGMENT
Medium Frequency

BEST UNIVERSAL PERK
Flight Discount Voucher

Perk Preference by Age Segment

40-49 customers represent the highest-value segment



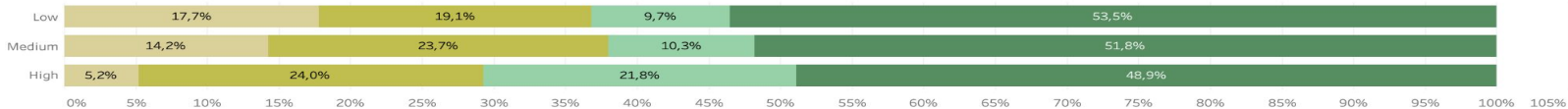
Perk Preference Intensity by Travel Frequency

Medium-frequency travelers show the strongest overall engagement

	Flight Discount Voucher	Free Checked Bag	Hotel Discount Voucher	Hotel Night Discount
Low	10.4K	1.5K	3.9K	4.2K
Medium	11.5K	2.4K	5.1K	2.6K
High	4.1K	2.1K	2.0K	257

Perk Preference Share by Hotel Activity

High hotel users show the strongest preference for Free Checked Bag (21.84%)



Segment Summary

Most Valuable Segment:
40-49

Most Engaged Travelers:
Medium Frequency

Best Universal Perk:
Flight Discount Voucher

Strategic Insights

1. Prioritize customers aged 40-49 for loyalty campaigns.
2. Standardize Flight Discount Voucher as the primary loyalty incentive.
3. Promote Free Checked Bag offers to high hotel users.
4. Focus acquisition and retention efforts on medium-frequency travelers.

Customers aged 40-49 and medium-frequency travelers represent the highest-value engagement opportunities.

Strategic Loyalty Recommendations Dashboard



Strategic Loyalty Recommendations

Recommended customer perks and targeting strategy based on behavioral analysis

PRIMARY PERK
Most preferred loyalty reward across all customers
Flight Discount Voucher

BEST UPSELL TARGET
Largest high-engagement customer segment
40-49
30.8% of engaged customers

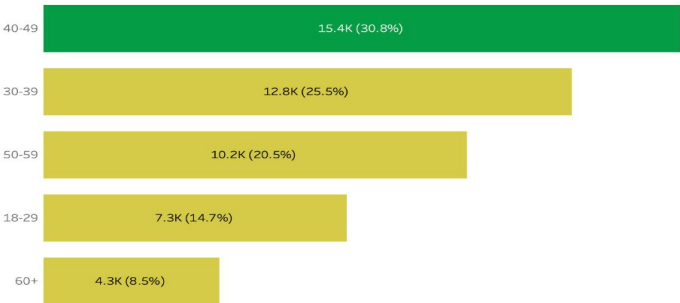
Perk Preference Matrix

Medium-frequency travelers show the strongest overall engagement, with Flight Discount Voucher consistently preferred across all segments.

Observed Perk Preference by Travel Frequency
Observed customer preference distribution across travel frequency segments

	Flight Discount Voucher	Free Checked Bag	Hotel Discount Voucher	Hotel Night Discount
Low	10.4K	1.5K	3.9K	4.2K
Medium	11.5K	2.4K	5.1K	2.6K
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Segment Opportunity Ranking
Ranked by total customers demonstrating loyalty perk engagement



Recommended Marketing Actions

- Standardize Flight Discount Voucher as the primary loyalty incentive.
- Prioritize customers aged 40-49, representing 30.8% of engaged users.
- Introduce targeted Free Checked Bag promotions for high-frequency travelers, where preference share is highest.
- Pilot Hotel Discount Voucher campaigns among medium-frequency travelers and measure conversion uplift.

Flight Discount Voucher should become the core loyalty incentive, supported by targeted campaigns for specific behavioral segments.

Recommendations & Next Steps



Recommendations

- ✓ Standardize Flight Discount Voucher
- ✓ Prioritize 40–49 customer segment
- ✓ Focus retention on medium-frequency travelers
- ✓ Promote Free Checked Bag offers to high hotel-usage customers

Future Opportunities

- ✓ A/B testing
- ✓ Churn prediction
- ✓ Machine learning personalization
- ✓ Dynamic perk recommendation engine

Thank You



Questions?

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